

Practice Quiz: Digital Transformation — Module 06: Learning

Name: ____ Date: ____

Part A: Multiple Choice

1. Zillow's iBuying failure was caused by: A) Bad management B) Model drift — the ML model was trained on historical patterns that stopped holding in a rapidly changing market C) Competition D) Regulatory issues
 2. **Model drift** occurs when: A) Models move B) The statistical relationship between inputs and outputs changes over time, degrading model performance C) Data is lost D) Hardware fails
 3. The ML model lifecycle requires: A) Train once, deploy forever B) Continuous monitoring and periodic retraining to maintain model accuracy C) No monitoring D) Daily retraining
 4. **MLOps** addresses: A) Marketing operations B) The operational infrastructure for deploying, monitoring, and maintaining ML models at scale C) Office operations D) Military operations
 5. "Hidden technical debt in ML systems" means: A) Budget overruns B) ML systems accumulate maintenance burdens (data dependencies, feedback loops, configuration complexity) that are invisible until they cause failures C) Developer debt D) Infrastructure costs
 6. **Supervised learning** requires: A) A supervisor B) Labeled data — examples where both inputs and correct outputs are known — to train the model C) More computing power D) Real-time data
 7. **Organizational ML maturity** progresses from: A) Big to small B) Ad hoc (individual models) → repeatable (standard processes) → managed (MLOps infra) → optimized (embedded in business) C) Simple to complex D) Fast to slow
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Part B: Short Analysis

1. Your organization's customer churn model has been in production for 2 years without retraining. Design a monitoring system that detects drift and triggers retraining before accuracy degrades significantly.
2. Your data science team builds great models but deployment regularly fails. Diagnose the likely organizational and technical gaps and design an MLOps improvement plan.
3. A recommendation model creates a feedback loop: it recommends popular items, users click on them, which makes them more popular, narrowing recommendation diversity. Analyze this as an Active Inference problem and propose a solution.